Hákon Freyr Gunnarsson

AI engineer

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PROFILE

AI engineer with a mathematics and statistics foundation (BSc mathematics, MSc AI and data science, two years as a statistician at deCODE genetics) and production systems shipped in genomics, finance, travel and enterprise software. From March to October 2026 I ran a build sprint at my own company, Krates, taking agent-first back-office software and the AI engineering team that develops it to an independently verified working core. That system now largely runs on its own, and I'm returning to full-time engineering to build AI systems that can be trusted with real work. I trust measurements over demos.

EXPERIENCE

Founder & Lead Engineer · Krates ehf.

Mar 2026 – present

A seven-month build sprint taking agent-first back-office software from design to a verified core. Krates continues as a side company, with development carried mainly by the agent team.

- **Krepis:** seven event-sourced, idempotent kernels (accounting, payroll, workforce, payments, inventory and more) with a dry run on every action. Business rules are Logica laws compiled to DuckDB SQL that return violation sets. hakonfreyr.com/projects/krepis
- **Verified unsupervised:** AI agents ran a fictional 12-employee company's month through six kernels. An independent checker found zero differences across 4,913 comparisons, and payroll was exact to the króna.
- **Sókrates:** an on-premises NixOS appliance for agent back-office work. Execution is crash-safe (DBOS), and every external action carries a signed consent token and an audit trail. hakonfreyr.com/projects/sokrates
- **The Weave:** designed and led a five-agent AI engineering team: 106 merged changes a week (July to mid-September 2026), 94% independently reviewed, 0.14% reverted, 250 ISK ($2) per change on average. hakonfreyr.com/projects/the-weave

Backend Engineer · Wise lausnir (Microsoft Dynamics partner)

Oct 2024 – Mar 2026

- Designed and deployed Grimoire, an internal knowledge platform on Azure (Container Apps, Bicep IaC, Neo4j AuraDB) that Claude uses as an MCP server, rolled out to every technical employee through Claude Team admin. A full OAuth 2.0 stack (RFC 7591, 8414, 9728) lets colleagues sign in with their company email and nothing to install. hakonfreyr.com/projects/grimoire
- Cut municipal onboarding from legacy on-premises systems into Fabric/OneLake from about three weeks to half a workday. Built the self-healing ETL framework for the Business Central solution using PySpark, a medallion architecture and models generated from OpenAPI schemas.

Data Scientist & Business Intelligence Lead · Travelshift

Aug 2022 – Sep 2024

Rebuilt the data infrastructure and the financial automation of Iceland's largest travel marketplace.

- **Forecasting:** a cash-flow forecasting model with a **0.42% margin of error** on 2.4B ISK of volume over a six-month horizon.
- **Reconciliation:** automated payment reconciliation across Adyen, Teya and internal systems. Unreconcilable transactions fell from **1 in 35 to under 1 in 1,000**.
- **Data platform:** a Kimball star schema in Snowflake, ETL across internal and external systems, and Power BI reporting.
- **Operations:** reduced the booking department from seven contractors to two while improving its performance. Started with no finance background and within months was producing annual reports that passed audit.

Data Analyst & Product Owner · Alfreð Atvinnuleit

May 2021 – Aug 2022

Product owner for the Giggó gig-economy platform (system architecture, data model). Built the BI environment with real-time dashboards. Wrote the grant proposal that secured 30M ISK from Rannís.

Instructor, Department of Computer Science · Reykjavík University

2018 – 2021

Programming, Data Structures, Calculus & Statistics, and Discrete Mathematics II, taught alongside the MSc.

Statistician & Bioinformatician · deCODE genetics (Amgen)

2015 – 2017

Designed a haplotype compression algorithm with more than 10× the compression of gzip, which let whole chromosomes be loaded into memory instead of in segments. Large-scale genetic data analysis.

SELECTED PROJECTS

Homegrown Hero Films · homegrownherofilms.com

AI video · GPU pipelines · YouTube & TikTok · 2026

Short films where a child is the hero. Behind them sits a video pipeline on rented GPUs, with my own open-source extensions to the MiniMax H3 video model, AI agents that review every film before release, upscaling and frame-interpolation finishing, and scheduled publishing.

Residual Modeling

Python · LightGBM · Postgres/PostGIS · FastAPI · Next.js · 2026

Icelandic home valuation that splits each sale price into multiplicative layers (structure, location, local market, liquidity and timing), fitted as residuals in log-price space, so every estimate shows where its value comes from. On 12,709 time-ordered out-of-fold sales the median error is 6.6%, and 5.9% for apartments.

hakonfreyr.com/projects/residual-modeling

Autopod

Go · Python/FastAPI · SQLite · Cloudflare R2 · speech-to-text · LLMs

Drop in a video and it is published to YouTube, Spotify and Apple Podcasts with subtitles in three languages, a summary and clips of the most quotable moments. My first Go project, shipped to multi-tenant production. hakonfreyr.com/projects/autopod

SKILLS

AI & agents	pydantic-ai, MCP, Anthropic/OpenAI/Gemini APIs, agent evaluation, RAG and embeddings, durable agent workflows, tracing with Logfire/OpenTelemetry, local inference, diffusion models
Languages	Python (expert), SQL, TypeScript, Go, R, Logica
Data	Postgres, DuckDB/DuckLake, Snowflake, Neo4j, Fabric/PySpark, Kimball modelling, event sourcing, Power BI
Infrastructure	NixOS, Docker, Azure (Container Apps, Bicep), Cloudflare, Tailscale, GitHub Actions
Engineering	Domain-driven design, architecture enforced in CI (import-linter, tach), typed errors, uv/ruff/ty
Spoken	Icelandic (native), English (fluent)

EDUCATION

MSc Computer Science · Reykjavík University — AI and data science; thesis on recommendation systems 2017 – 2021

BSc Mathematics · University of Iceland — computational mathematics and computer science 2011 – 2016